

# Vishwavrinda Ayurveda API And System Documentation

This document describes the backend software, API contract, authentication, data flow, and important file locations for the Vishwavrinda Ayurveda WhatsApp AI admin system.

## Application Layout

```
auryvedic/  
backend/  
src/  
constants/ API paths, auth constants, pagination, database names, indexes  
lib/ shared helpers for pagination, ObjectId, regex, body validation  
parsers/ WhatsApp webhook payload parsing  
plugins/ Fastify plugins such as JWT authentication  
routes/ HTTP route handlers  
services/ MongoDB, OpenAI, WhatsApp, appointment, dashboard, knowledge logic  
types/ TypeScript models and Fastify/JWT augmentation  
docs/ API and system documentation  
scripts/ demo seed and PDF generation scripts  
knowledge.md file-based clinic knowledge used by the AI  
frontend/  
app/ Next.js dashboard pages  
components/ shadcn-style UI components  
lib/ API client, auth helpers, formatting  
proxy.ts frontend route protection
```

## Authentication

Admin dashboard APIs are protected with JWT.

Public backend routes:

- 'GET /'
- 'GET /webhook'
- 'POST /webhook'
- 'POST /api/auth/login'
- 'OPTIONS \*'

Protected backend routes:

- 'GET /test-whatsapp'
- 'GET /test-ai'
- 'GET /api/dashboard/stats'
- 'GET /api/appointments'
- 'PATCH /api/appointments/:id'
- 'GET /api/users'
- 'GET /api/users/:phoneNumber/messages'
- 'GET /api/knowledge'
- 'PUT /api/knowledge'

Admin credentials:

```
username: Test
password: 1247
role: admin
```

Authenticated requests must include:

```
Authorization: Bearer <jwt>
```

Relevant files:

- 'backend/src/plugins/auth.ts'
- 'backend/src/routes/auth.ts'
- 'backend/src/constants/auth.ts'
- 'backend/src/types/auth.ts'
- 'frontend/app/login/page.tsx'
- 'frontend/proxy.ts'
- 'frontend/lib/api-client.ts'

## System Flow

```
Patient on WhatsApp
|
v
Meta WhatsApp Cloud API
|
v
POST /webhook
|
v
backend/src/parsers/whatsapp.ts
|
v
backend/src/services/conversation.ts
|
+--> upsertUser(phoneNumber)
| |
| v
| MongoDB users
|
+--> createMessage(role=user)
| |
| v
| MongoDB messages
|
+--> getKnowledge()
| |
| v
| backend/knowledge.md
|
+--> generateReply()
| |
| v
| OpenAI
|
+--> AIResult type=message
```

```

| |
| v
| sendAndStoreMessage()
| |
| +--> sendMessages() to WhatsApp
| |
| +--> createMessage(role=assistant) only after successful send
|
+--> AIResult type=appointment
|
v
createAppointment()
|
v
MongoDB appointments

```

## Admin Dashboard Flow

```

Admin Browser
|
v
frontend/proxy.ts checks JWT cookie
|
+--> no valid token -> /login
|
+--> valid admin token -> dashboard page
|
v
frontend/lib/api-client.ts adds Authorization header
|
v
Fastify preHandler verifies JWT and role=admin
|
v
backend route handler

```

## Data Stores

MongoDB collections:

- 'users'
- 'messages'
- 'appointments'

File storage:

- 'backend/knowledge.md'

Indexes created on startup:

- 'users.phoneNumber' unique
- 'users.lastActiveAt'
- 'users.firstSeenAt'
- 'messages.phoneNumber + createdAt'
- 'appointments.createdAt'

- 'appointments.isCompleted + createdAt'
- 'appointments.phoneNumber'

## Common Pagination

Query parameters:

- 'page': positive integer, default '1'
- 'limit': positive integer, default '20', maximum '100'

Response shape:

```
{
  "data": [],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 0,
    "totalPages": 0
  }
}
```

## API Reference

### GET /

File:

- 'backend/src/index.ts'

Auth:

- Public

Purpose:

- Health check endpoint.

Parameters:

- None

Returns:

```
{
  "status": "ok",
  "service": "WhatsApp AI Bot"
}
```

### POST /api/auth/login

Files:

- 'backend/src/routes/auth.ts'
- 'backend/src/plugins/auth.ts'

Auth:

- Public

Purpose:

- Login for the single hardcoded admin user.

Body:

```
{
  "username": "Test",
  "password": "1247"
}
```

Returns:

```
{
  "token": "jwt-token",
  "user": {
    "username": "Test",
    "role": "admin"
  }
}
```

Errors:

- '401' when username or password is incorrect.

## **GET /webhook**

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/config.ts'

Auth:

- Public

Purpose:

- Meta WhatsApp webhook verification.

Query parameters:

- 'hub.mode': expected 'subscribe'
- 'hub.verify\_token': must match 'VERIFY\_TOKEN'
- 'hub.challenge': challenge string returned to Meta

Returns:

- '200' with challenge string when valid.
- '403' with 'Verification failed' when invalid.

## **POST /webhook**

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/parsers/whatsapp.ts'
- 'backend/src/services/conversation.ts'

Auth:

- Public

Purpose:

- Receives WhatsApp events, stores messages, calls the AI, sends replies, and creates appointments when the AI returns complete appointment data.

Body:

- Meta WhatsApp webhook payload.

Returns:

```
{  
  "success": true  
}
```

Side effects:

- Upserts user in 'users'.
- Stores incoming user message in 'messages'.
- Reads 'backend/knowledge.md'.
- Calls OpenAI.
- Sends WhatsApp message.
- Stores assistant reply only after WhatsApp send succeeds.
- Creates appointment in 'appointments' when needed.

## **GET /test-whatsapp**

File:

- 'backend/src/routes/webhook.ts'

Auth:

- Admin JWT required

Purpose:

- Sends a fixed WhatsApp test message.

Parameters:

- None

Returns:

```
{
  "success": true
}
```

## **GET /test-ai**

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/services/knowledge.ts'
- 'backend/src/services/openai.ts'

Auth:

- Admin JWT required

Purpose:

- Loads knowledge and sends a fixed test question to OpenAI.

Parameters:

- None

Returns:

```
{
  "success": true,
  "reply": {
    "type": "message",
    "reply": "string",
    "appointment": null
  }
}
```

## **GET /api/dashboard/stats**

Files:

- 'backend/src/routes/dashboard.ts'
- 'backend/src/services/dashboard.ts'

Auth:

- Admin JWT required

Purpose:

- Returns counters and chart trend data for the dashboard.

Query parameters:

- 'period': optional 'week' or 'month', default 'week'
- 'anchorDate': optional date in 'YYYY-MM-DD'; when omitted, today is used

Returns:

```
{
  "totalUsers": 10,
  "newUsersThisWeek": 4,
  "newUsersThisMonth": 8,
  "totalMessages": 40,
  "totalAppointments": 10,
  "pendingAppointments": 6,
  "completedAppointments": 4,
  "appointmentsThisWeek": 3,
  "appointmentsThisMonth": 9,
  "trend": {
    "period": "week",
    "anchorDate": "2026-07-09",
    "startDate": "2026-07-03",
    "endDate": "2026-07-09",
    "points": [
      {
        "date": "2026-07-03",
        "label": "3 Jul",
        "users": 5,
        "appointments": 2
      }
    ]
  }
}
```

Errors:

- '400' for invalid period.
- '400' for invalid anchor date.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

## GET /api/appointments

Files:

- 'backend/src/routes/appointments.ts'
- 'backend/src/services/appointments.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated appointment requests.



Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'
- 'status': optional 'pending', 'done', or 'all', default 'all'
- 'search': optional search across 'patientName' and 'phoneNumber'

Returns:

```
{
  "data": [
    {
      "_id": "64f000000000000000000001",
      "patientName": "Aarav Sharma",
      "phoneNumber": "+919880000001",
      "preferredDate": "2026-07-10",
      "preferredTime": "10:30 AM",
      "reason": "Digestive discomfort",
      "preferredContactMethod": "whatsapp",
      "source": "whatsapp",
      "isCompleted": false,
      "createdAt": "2026-07-09T10:00:00.000Z",
      "updatedAt": "2026-07-09T10:00:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '400' for invalid status.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

## **PATCH /api/appointments/:id**

Files:

- 'backend/src/routes/appointments.ts'
- 'backend/src/services/appointments.ts'

Auth:

- Admin JWT required

Purpose:

- Updates only appointment completion status.

Path parameters:

- 'id': MongoDB ObjectId string

Body:

```
{
  "isCompleted": true
}
```

Returns:

```
{
  "success": true,
  "data": {
    "_id": "64f000000000000000000001",
    "patientName": "Aarav Sharma",
    "phoneNumber": "+919880000001",
    "preferredDate": "2026-07-10",
    "preferredTime": "10:30 AM",
    "reason": "Digestive discomfort",
    "preferredContactMethod": "whatsapp",
    "source": "whatsapp",
    "isCompleted": true,
    "createdAt": "2026-07-09T10:00:00.000Z",
    "updatedAt": "2026-07-09T10:15:00.000Z",
    "completedAt": "2026-07-09T10:15:00.000Z"
  }
}
```

Rules:

- Only 'isCompleted' can be changed.
- Extra body fields are rejected.
- Setting 'isCompleted=true' adds 'completedAt'.
- Setting 'isCompleted=false' removes 'completedAt'.

Errors:

- '400' for invalid ObjectId.
- '400' for invalid body.
- '401' when JWT is missing or invalid.
- '404' when appointment is not found.
- '500' for unexpected database errors.

## GET /api/users

Files:

- 'backend/src/routes/users.ts'
- 'backend/src/services/users.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated WhatsApp users.

Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'
- 'search': optional phone number search

Returns:

```
{
  "data": [
    {
      "phoneNumber": "+919880000001",
      "firstSeenAt": "2026-07-01T10:00:00.000Z",
      "lastActiveAt": "2026-07-09T10:00:00.000Z",
      "totalMessages": 4,
      "createdAt": "2026-07-01T10:00:00.000Z",
      "updatedAt": "2026-07-09T10:00:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

## **GET /api/users/:phoneNumber/messages**

Files:

- 'backend/src/routes/users.ts'
- 'backend/src/services/messages.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated conversation history for one WhatsApp phone number.

Path parameters:

- 'phoneNumber': WhatsApp phone number

Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'

Returns:

```
{
  "data": [
    {
      "phoneNumber": "+919880000001",
      "role": "user",
      "content": "Namaste, I need an appointment.",
      "createdAt": "2026-07-09T10:00:00.000Z"
    },
    {
      "phoneNumber": "+919880000001",
      "role": "assistant",
      "content": "Please share your preferred date and time.",
      "createdAt": "2026-07-09T10:01:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 2,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '400' for invalid phone number.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

## GET /api/knowledge

Files:

- 'backend/src/routes/knowledge.ts'
- 'backend/src/services/knowledge.ts'

Auth:

- Admin JWT required

Purpose:

- Reads file-based clinic knowledge.

Parameters:

- None

Returns:

```
{
  "content": "markdown content"
}
```

Errors:

- '401' when JWT is missing or invalid.
- '500' for filesystem errors.

## **PUT /api/knowledge**

Files:

- 'backend/src/routes/knowledge.ts'
- 'backend/src/services/knowledge.ts'

Auth:

- Admin JWT required

Purpose:

- Replaces 'backend/knowledge.md'.

Body:

```
{
  "content": "updated markdown content"
}
```

Validation:

- 'content' must be a string.
- 'content' must not be empty after trimming.
- 'content' maximum length is '100000' characters.

Returns:

```
{
  "success": true
}
```

Errors:

- '400' for invalid content.
- '401' when JWT is missing or invalid.
- '500' for filesystem errors.

## **Important Files By Responsibility**

- Backend entrypoint: 'backend/src/index.ts'
- Route registration: 'backend/src/routes/index.ts'
- Auth plugin: 'backend/src/plugins/auth.ts'

- API path constants: 'backend/src/constants/api.ts'
- Database constants: 'backend/src/constants/database.ts'
- Pagination helper: 'backend/src/lib/pagination.ts'
- Request body validators: 'backend/src/lib/request-body.ts'
- WhatsApp parser: 'backend/src/parsers/whatsapp.ts'
- WhatsApp sender: 'backend/src/services/whatsapp.ts'
- OpenAI service: 'backend/src/services/openai.ts'
- Conversation orchestrator: 'backend/src/services/conversation.ts'
- Appointment service: 'backend/src/services/appointments.ts'
- User service: 'backend/src/services/users.ts'
- Message service: 'backend/src/services/messages.ts'
- Dashboard service: 'backend/src/services/dashboard.ts'
- Knowledge service: 'backend/src/services/knowledge.ts'
- Frontend API client: 'frontend/lib/api-client.ts'
- Frontend login page: 'frontend/app/login/page.tsx'
- Frontend protected route proxy: 'frontend/proxy.ts'